Light

GCSE Computer Science — Cyber Security Threats (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A type of malware secretly monitors a user's activity and sends information to a third party without consent.
What is this malware called?

- ○ **A.** Spyware
- ○ **B.** Firmware
- ○ **C.** Middleware
- ○ **D.** Groupware

Q2 (1 mark)

An attacker uses software to systematically try every possible password combination until the correct one is found.
What type of attack is this?

- ○ **A.** A phishing attack
- ○ **B.** A brute force attack
- ○ **C.** A denial of service attack
- ○ **D.** A man-in-the-middle attack

Q3 (1 mark)

Multiple computers flood a website with requests, making it unavailable to legitimate users.
What type of attack is this?

- ○ **A.** SQL injection
- ○ **B.** Phishing
- ○ **C.** Distributed denial of service (DDoS)
- ○ **D.** Ransomware

Q4 (1 mark)

An attacker watches over someone's shoulder as they enter their PIN at a cash machine.
What is this technique called?

- ○ **A.** Phishing
- ○ **B.** Blagging
- ○ **C.** Pharming
- ○ **D.** Shoulder surfing

Q5 (1 mark)

An attacker pretends to be a delivery driver and convinces a receptionist to reveal employee names and email addresses.
What is this technique called?

- ○ **A.** Blagging
- ○ **B.** Shoulder surfing
- ○ **C.** Brute force
- ○ **D.** SQL injection

Q6 (1 mark)

A user types their bank's correct web address but is secretly redirected to a fake website that looks identical.
What type of attack is this?

- ○ **A.** Shoulder surfing
- ○ **B.** Pharming
- ○ **C.** Blagging
- ○ **D.** Brute force

Q7 (1 mark)

An attacker enters malicious code into a website's login form to access the database without valid credentials.
What type of attack is this?

- ○ **A.** Phishing

- ○ **B.** Denial of service
- ○ **C.** SQL injection
- ○ **D.** Shoulder surfing

Q8 (1 mark)

A company requires passwords to contain at least 12 characters including uppercase, lowercase, numbers, and symbols.
What is this an example of?

- ○ **A.** Encryption
- ○ **B.** Two-factor authentication
- ○ **C.** Biometric security
- ○ **D.** A password policy

Q9 (1 mark)

A user enters their password and then must also enter a code sent to their mobile phone.
What security method is this?

- ○ **A.** Two-factor authentication
- ○ **B.** Single sign-on
- ○ **C.** Biometric authentication
- ○ **D.** Password hashing

Q10 (1 mark)

A smartphone unlocks when it recognises the owner's fingerprint.
What type of security is this?

- ○ **A.** Password authentication
- ○ **B.** Biometric authentication
- ○ **C.** Network authentication
- ○ **D.** Token authentication

Q11 (1 mark)

Software installed on a computer displays unwanted advertisements and pop-ups.
What type of software is this?

- ○ **A.** Ransomware
- ○ **B.** A firewall
- ○ **C.** Adware
- ○ **D.** Firmware

Q12 (1 mark)

A computer automatically downloads and installs security patches from the operating system manufacturer.
Why is this important for security?

- ○ **A.** It makes the computer run faster
- ○ **B.** It increases the computer's storage space

- ○ **C.** It removes all existing malware
- ○ **D.** It fixes known vulnerabilities that attackers could exploit

Q13 (2 marks)

A website locks user accounts after five failed login attempts.
What type of attack does this measure primarily protect against?

- ○ **A.** Brute force attacks, because attackers cannot try unlimited password combinations
- ○ **B.** Phishing attacks, because users cannot enter passwords on fake sites
- ○ **C.** SQL injection attacks, because malicious code cannot be entered
- ○ **D.** DDoS attacks, because traffic is limited to five requests

Q14 (2 marks)

A student explains the difference between phishing and pharming.
Which statement is correct?

- ○ **A.** Phishing encrypts files; pharming demands ransom payments
- ○ **B.** Phishing requires users to click malicious links; pharming redirects users automatically even when typing correct addresses
- ○ **C.** Phishing targets databases; pharming targets network traffic
- ○ **D.** Phishing uses physical observation; pharming uses telephone calls

Q15 (2 marks)

An online retailer experiences a DDoS attack during a major sales event. What would be the most likely impact on the business?

- ○ **A.** Customer data would be encrypted and held for ransom
- ○ **B.** Customer passwords would be stolen through fake login pages
- ○ **C.** The website would become unavailable, causing lost sales and damage to reputation
- ○ **D.** The company's internal emails would be intercepted

Q16 (2 marks)

A web developer wants to protect their website from SQL injection attacks.
Which measure would be most effective?

- ○ **A.** Installing a stronger firewall to block all database traffic
- ○ **B.** Requiring users to change their passwords every 30 days
- ○ **C.** Adding CAPTCHA tests to all forms on the website
- ○ **D.** Validating and sanitising all user input before processing it in database queries

Q17 (2 marks)

A bank introduces two-factor authentication for online banking.
Why does this significantly improve security compared to passwords alone?

- ○ **A.** An attacker who steals the password still cannot access the account without the second factor

- ○ **B.** It makes passwords longer and more complex automatically
- ○ **C.** It encrypts all data transmitted between the user and the bank
- ○ **D.** It prevents all types of malware from infecting the user's computer

Q18 (2 marks)

A company considers using fingerprint scanners instead of passwords for building access.
What is one advantage and one disadvantage of biometric authentication?

- ○ **A.** Advantage: fingerprints can be easily changed if compromised; Disadvantage: fingerprints are easy to forget
- ○ **B.** Advantage: fingerprints cannot be forgotten or shared; Disadvantage: if compromised, fingerprints cannot be changed like passwords
- ○ **C.** Advantage: fingerprints work over the internet; Disadvantage: fingerprints require electricity
- ○ **D.** Advantage: fingerprints are free to implement; Disadvantage: fingerprints only work in daylight

Q19 (2 marks)

Malware infects thousands of computers and allows an attacker to control them remotely as a group.
What is this network of infected computers called?

- ○ **A.** A firewall cluster
- ○ **B.** A virtual private network

- ○ **C.** A botnet
- ○ **D.** A local area network

Q20 (2 marks)

A company's network policy requires all employees to use a VPN when working remotely.
What security benefit does this provide?

- ○ **A.** It prevents employees from accessing any websites
- ○ **B.** It automatically removes all malware from employee devices
- ○ **C.** It increases the internet connection speed for remote workers
- ○ **D.** It encrypts data transmitted between the employee's device and the company network, protecting it from interception

Q21 (2 marks)

An employee receives a phone call from someone claiming to be from Microsoft, saying their computer has a virus and asking for remote access.
What should the employee do?

- ○ **A.** Refuse access, end the call, and report it to IT security as a likely blagging attempt
- ○ **B.** Provide remote access immediately to fix the virus
- ○ **C.** Give the caller their password so they can check the computer
- ○ **D.** Transfer the call to a colleague to handle

Q22 (2 marks)

A company replaces old computers that contained confidential customer data.
What should be done with the old hard drives to protect data security?

- ○ **A.** Simply delete all files before disposing of the computers
- ○ **B.** Physically destroy or securely wipe the drives to prevent data recovery
- ○ **C.** Donate the computers to charity with the drives intact
- ○ **D.** Store the old computers in a cupboard indefinitely

Q23 (2 marks)

A type of malware hides deep in the operating system and is designed to avoid detection by anti-malware software.
What is this type of malware called?

- ○ **A.** Adware
- ○ **B.** A cookie
- ○ **C.** Shareware
- ○ **D.** A rootkit

Q24 (3 marks)

A cyber attack on a company unfolds in stages:
• Stage 1: Attackers send emails with malicious attachments to employees
• Stage 2: When opened, malware installs giving attackers remote access
• Stage 3: Attackers move through the network, gaining higher privileges
• Stage 4: Sensitive data is copied and sent to external servers

Which term describes this type of sophisticated, multi-stage attack?

- ○ **A.** An Advanced Persistent Threat (APT) using multiple techniques to achieve long-term access and data exfiltration
- ○ **B.** A brute force attack targeting password databases
- ○ **C.** A simple denial of service attack flooding the network
- ○ **D.** A single phishing email stealing one user's password

Q25 (3 marks)

A hospital is implementing a new authentication system for accessing patient records. They are evaluating three options:
• Option A: Password only
• Option B: Password plus fingerprint scan
• Option C: Fingerprint scan only
Which evaluation correctly assesses these options for a healthcare environment?

- ○ **A.** Option A is best because passwords are the most secure form of authentication
- ○ **B.** Option B provides the strongest security by combining something you know with something you are; Option A is vulnerable to password theft; Option C lacks a backup if biometrics fail
- ○ **C.** Option C is best because biometrics alone provide maximum security
- ○ **D.** All three options provide identical security levels

Q26 (3 marks)

A web application developer wants to protect against multiple attack types. They implement:

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• Input validation on all forms
• Parameterised database queries
• Rate limiting on login attempts
• HTTPS encryption
```

Which attack types do these measures protect against?

- ○ **A.** All four measures prevent DDoS attacks only
- ○ **B.** Input validation prevents ransomware; HTTPS prevents all malware
- ○ **C.** Input validation and parameterised queries prevent SQL injection; rate limiting prevents brute force attacks; HTTPS prevents eavesdropping on transmitted data
- ○ **D.** These measures provide no security benefits for web applications

Q27 (3 marks)

A company conducts a simulated phishing test. Results show:
• 45% of employees clicked the malicious link
• 30% entered their credentials on the fake login page
• 5% reported the suspicious email to IT security
Which analysis of these results is correct?

- ○ **A.** The results are excellent; only 5% reported the email, which is above average
- ○ **B.** These results cannot be improved through training
- ○ **C.** The 30% credential submission rate is acceptable for most organisations
- ○ **D.** The high click rate (45%) and credential submission (30%) indicate significant vulnerability; the low reporting rate (5%) shows

employees do not know how to respond; comprehensive security awareness training is needed

Q28 (3 marks)

A school's network becomes infected with malware. Investigation reveals:
• A student inserted a USB drive found in the car park
• The drive contained a file named 'Exam_Answers.pdf.exe'
• The file installed malware when opened
• The malware spread to other computers on the network
Which combination of attack techniques and vulnerabilities were exploited?

- ○ **A.** Social engineering (baiting with enticing filename), Trojan delivery (disguised executable), and network propagation exploiting shared resources
- ○ **B.** SQL injection through the school website, combined with weak database security
- ○ **C.** Brute force attack on the school's WiFi password
- ○ **D.** Denial of service attack from external hackers

Q29 (3 marks)

A company discovers their customer database has been breached. The IT team proposes this response:
• Step 1: Immediately shut down all systems

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• Step 2: Delete all log files to hide the breach
• Step 3: Resume normal operations without notifying customers
• Step 4: Hope the attackers do not return
```

Which evaluation of this response plan is correct?

- ○ **A.** This is an excellent incident response plan following industry best practices
- ○ **B.** This plan is fundamentally flawed: evidence should be preserved not deleted; affected customers must be notified under data protection law; root cause analysis is needed to prevent recurrence; hoping attackers do not return is not a security strategy
- ○ **C.** Deleting log files is the correct first step in any breach response
- ○ **D.** Customer notification is optional and should be avoided to protect reputation

Q30 (3 marks)

A small business owner says: 'We don't need cyber security measures because hackers only target large companies.'
Which response correctly addresses this misconception?

- ○ **A.** This is correct; small businesses are never targeted by cyber attacks
- ○ **B.** Cyber security is only necessary for businesses that operate online
- ○ **C.** Small businesses are frequently targeted because they often have weaker security than large companies; automated attacks do not discriminate by company size; a breach could devastate a small business that lacks resources to recover
- ○ **D.** Antivirus software alone provides complete protection for small businesses

Total: **55** marks • Time: **30 min**

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